

Learning Generalizable Dexterous Grasping for Rigid Objects

Project Group 40: CAO GUIPU (A0351781B), HE CHENYU (A0335875N), YU MIAOYU (A0355875L)

Why:

- Robots in warehouses, homes, and manufacturing must grasp objects of varying shapes, masses, and frictions. Parallel-jaw grippers are efficient for regular items but offer limited contact options for irregular or task-relevant objects. An xArm7 with a 16-DoF LEAP Hand enables richer multi-finger contacts, yet coordinating its many joints is challenging. A grasping policy must also adapt its wrist and finger motions to different object geometries, rather than rely on a grasp learned for one object. Open-loop execution may also fail under pose errors or slip, motivating closed-loop adjustment during grasping.
- We therefore aim to learn a shared closed-loop policy that grasps diverse rigid objects and generalises to unseen objects, coordinating wrist and finger motions during approach, grasp, lift, and hold. The project focuses on grasping one rigid object at a time on a table (no clutter), using Isaac Lab with diverse YCB objects. We exclude camera occlusion and sim-to-real transfer to isolate the core challenges of high-dimensional contact control and generalization to unseen objects.

Conventional Algorithms

- **Analytic planners** (e.g., GraspIt!) search for force-closure grasps using object geometry and friction models. They offer interpretable grasp quality measures, but planning can be expensive for high-DoF hands, and open-loop execution of the selected poses cannot adjust to pose errors or slip.
- **Data-driven methods** (e.g., Dex-Net 2.0 for parallel-jaw grippers) predict grasp quality from depth images and can generalise to novel objects. However, selecting a grasp does not by itself determine how wrist and finger motions should adapt during execution.
- **Both approaches** can be combined with feedback control. Unlike open-loop execution, our closed-loop policy continuously adjusts wrist and finger motions from observed states. We study whether a shared RL policy can learn from interaction with diverse objects to achieve stable lift-and-hold outcomes on both training and unseen objects.

Problem Statement:

- One xArm7 and LEAP Hand must use a shared policy to grasp rigid YCB objects of different shapes and sizes placed on a table, including unseen objects without further training. The hand starts open without contact. Within a 10 s episode, the robot must establish multi-finger contact, lift the object's lowest point at least 8 cm above the table, bring it within 5 cm of a target about 18 cm above its initial centre, and maintain a stable hold for 2 s.
- The policy receives a 256-point object surface cloud in the palm frame, robot proprioception, the relative object and goal state, the previous action, and task progress. These quantities are obtained from simulation. The training critic may additionally use mass, friction, contact forces, object clearance, and table contact; these privileged quantities are not required by the execution policy.
- At 30 Hz, the agent commands bounded wrist translation/rotation increments and 16 finger-joint increments. Damped least-squares IK converts wrist commands to arm joint targets, and position-PD controllers execute at 120 Hz. The object moves only through contact, gravity, friction, and inertia.

RL Cast:

- **Observation space (low dim + point cloud):**
 - Robot joint state (46): positions/velocities of 7 arm + 16 hand joints.

- Palm state (15): position, 6D orientation, linear/angular velocity.
- Fingertip state (12): positions of four fingertips in palm frame.
- Object state (15): relative position, 6D orientation, relative velocities.
- Goal state (3): goal position relative to palm.
- Object size (3): axis-aligned dimensions.
- Previous action (22): last wrist and finger command.
- Task progress (3): remaining time, hold progress, lifted flag.
- Point cloud (256×3): surface points in palm frame, encoded separately.
- **Action space (22-dim continuous):** wrist translation (3), wrist rotation (3), finger joints (16) – bounded deltas (~0.01 m, 0.05 rad, 0.05 rad per step). Sampled from diagonal Gaussian during training.
- **Reward structure:**
 - Progress reward for hand approaching object surface ($+0.5 \times$ change in reach potential).
 - Small reward for distinct finger contacts (up to +0.2) to encourage multi finger use.
 - Larger rewards for lift progress (+2) and goal approach (+1), enabled only after at least two fingers contact.
 - Hold reward (+2/s) and a one time success bonus (+10) after 60 stable steps.
 - Small penalties for large actions (-0.05) and high joint speed (-0.01); larger penalties for forbidden collisions and dropping (-2).
- **Neural Network Structure:** A point-cloud encoder (shared MLP per point, max-pool) extracts a geometry feature, concatenated with the low-dim state, then passed through an Actor-Critic MLP. The actor outputs the mean of 22 continuous actions (and learned noise); the critic outputs a value and may use privileged info. Exact layer sizes chosen during initial gr-asp-learning experiments.

RL algorithms and envisioned results:

- We will use Proximal Policy Optimization (PPO) with Generalized Advantage Estimation. PPO supports the 22-dimensional continuous action space, works well with parallel simulation, and limits overly large updates via its clipped objective. An existing single-object, state-only PPO pipeline has passed short execution checks and provides a starting point for learning a shared policy across rigid objects.
- DQN is not selected due to the impracticality of discretising 22 continuous actions. Soft Actor-Critic may be explored if time permits, but PPO remains the primary method for easier integration with our pipeline.
- Training starts with the YCB tomato-soup can, then extends to a shared policy for diverse rigid objects. Validation objects guide tuning, while final evaluation tests the frozen policy on training and held-out YCB objects without object-specific fine-tuning. Across training seeds, we report per-object and macro-average stable success rates, lift rate, post-lift retention, time to success, drop rate, and collision rate. Evaluation covers different object shapes and sizes, with pose observation noise or external pushes for robustness testing.
- The intended result is a shared dexterous grasping policy that can lift and hold diverse rigid objects and generalise to unseen objects in simulation. We will quantify the generalisation gap between training and held-out objects and analyse failure cases to identify the limits of the learned policy. We also expect the closed-loop policy to tolerate pose errors and disturbances better than the scripted open-loop reference. Sim-to-real transfer remains outside the project scope.